Bounded self-organising planning

Shared AI platform for FreshWay
Sub-assignment 5

Arrival-risk and unloading-time predictions supporting coordinated planning, with the Outbound planner approving every operational change.

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Research and implementation proposal. No platform deployment, pilot outcome or Board funding approval is claimed.

Contents and management summary

This report develops the Outbound part of FreshWay's integrated Smart Logistics vision. It combines the supplied team contributions, the corrected submitted SA3/SA4 work, the course requirements and a fresh check of the delivery data. The design remains a proposal.

Section	Page
1. Evidence and problem	3
2. Research and evidence maturity	4
3. Shared platform and departmental application	5
4. Coordination and adaptation	6
5. Decision rights and cross-delivery	7
6. Stakeholder impact map	8–9
7. KPIs and evaluation	10
8. Implementation, costs and risk	11
9. Evidence appendix and submission checks	12
10. AI declaration and responsibility	13
References	14–15

Management summary

The agreed direction is a company-wide AI platform. Outbound will use separately governed models for arrival risk and unloading duration. Its proposed trip agents will coordinate tentative planning options inside approved limits. The Outbound planner retains approval of every operational departure, route or stop-order change.

The historical data supports investigating ZE delivery reliability, but neither earlier scheduling nor AI has been shown to improve FreshWay performance. The next decision is to validate interfaces and demonstrate bounded coordination in shadow mode. Procurement and live operation follow only after requirements, costs, access, safety and evaluation criteria are agreed.

Prepared 4 November 2026. The team post's planned 2 November deadline has passed. The actual Brightspace deadline and submission status must be checked by the human team. No submission or recipient acceptance is claimed.

1 Evidence and problem

Outbound replenishes 120 stores from three DCs, including 22 stores in zero-emission zones (ZE). SA3 proposed window-aware planning and live ETA monitoring. Submitted SA4 added arrival-risk and unloading-time predictions with planner approval. SA5 develops how these services could coordinate and adapt on the shared platform (Company dossier, 2026, §§A2, A5; FreshWay Outbound Transport Team, 2026a, 2026b).

Table 1
Historical arrival measures and planning margins

Historical measure	ZE	Other stores
Valid deliveries	3,847	15,395
Supplied on_time flag	87.47%	98.38%
Inside both recorded bounds, inclusive	88.56%	94.08%
Appointment length	30 min	60 min
Planned arrival after opening	15 min	15 min
Remaining margin before closure	15 min	45 min

Note. Authors' calculations from FreshWay/store_replenishment.csv, fields ze_zone, on_time, actual_arrival, window_start, window_end and planned_arrival. 19,242 unique delivery IDs, 14 October 2024–11 January 2025. Percentages are arrival measures, not OTIF.

The team CSV contains 488 appended analysis/blank rows; only valid delivery records enter calculations. Fifty-eight supplied flags disagree with an inclusive closing-time rule, all exactly at the recorded closing minute. The inclusive-window measure additionally treats early arrivals as outside the window. A change of definition would change the percentage without changing any delivery. The data owner must confirm one rule before evaluation.

Interpretation and limits

ZE deliveries have less remaining scheduling margin. This association supports testing feasible earlier planning, but does not establish the cause of each late arrival. All appointments have a constant +15-minute planned offset; its operational reason remains unverified. Thirty- and sixty-minute appointments must also be distinguished from broader store acceptance hours and permit conditions.

Required quantities, accepted-delivery/POD records and actual departure times are missing. Full OTIF is therefore not calculable from this file. In 3,068 of 7,729 consecutive-stop transitions, recorded arrival precedes the minimum implied by previous arrival, unloading and the next recorded drive. These inconsistencies prevent a reliable causal route replay without data-owner clarification.

The department states OTIF 97.2% and a 98.5% company target. These are case references, separate from the calculated arrival percentages (Department file: Outbound transport, 2026, pp. 2–3).

2 Research and evidence maturity

Recent academic article

Gerrits, van Heeswijk, and Mes (2024) systematically review 169 transport-logistics studies selected from 553 search results. They organise self-organising logistics around delegated decisions and useful cooperation. Hybrid control can combine local choices with human oversight. Their review distinguishes agent-based decomposition of a central problem from genuine delegation and cooperation.

The evidence has limited operational maturity: 115 reviewed studies are classified at simulation maturity and only five as deployed. The search ended in May 2023. The review covers transport broadly and does not test grocery unloading, FreshWay's shared platform or its financial return. It supports a design framework and a testable hypothesis, rather than a benefit forecast (Gerrits et al., 2024, §§3–7).

Application to Outbound: our design inference

The shared server provides infrastructure. Arrival and unloading forecasts provide information. Outbound adds delegated choices only within a provisional planning workspace: trip agents choose candidate options, exchange feasibility and revise competing requests. A human still releases the operational plan. This is a proposed step toward partial self-organisation in planning; its implementation must demonstrate actual cooperation and adaptation.

The initial shadow test compares a central planner workflow against this bounded coordination mechanism using the same constraints. If the implementation only publishes forecasts or alerts and all planning decisions remain central, it must be described as predictive decision support.

Table 2
Evidence maturity and the claim supported

Evidence category	Assessment for this proposal
Tier 1	Requires accountable, independently examined production/outcome evidence under the course criteria. None establishes FreshWay's proposed system or benefits.
Tier 2	Requires traceable named and dated deployment evidence with corroboration. The academic review is not one such deployment case. No Tier 2 FreshWay deployment is claimed.
Tier 3	FreshWay's proposed concept is pre-pilot. A future demonstration or simulation can test a method but cannot establish production value.
Academic / internal evidence	The review supports concepts and research limits. Case documents define fictional case facts. Neither automatically becomes Tier 1 by being peer reviewed or internal.

Note. Course deployment-evidence criteria: Lesson 2 (2026), slides 49–50; Lesson 5 (2026), slides 17–19. Tier 4 covers untraceable promotional claims. Tiers assess the claim supported, separately from academic credibility.

3 Shared platform and Outbound application

The Outbound team has agreed the shared AI server/platform as its design direction. This does not establish Board funding, deployed infrastructure or acceptance by other departments. A shared logical platform can use managed hosting or physical servers; IT must compare deployment options after collecting requirements.

Table 3
Current and proposed architecture

Layer	Current case / proposed design
Source systems	Existing Routiq TMS, WMS–TMS API and vehicle board computers. Nightly BI supports analysis. Vendor-cloud position export requires a paid API, not yet verified or purchased.
Shared services: proposed	Identity/access control, departmental data areas, model registry, versioned deployment, event exchange, workload scheduling, monitoring, backup and recovery. All departments can state requirements; unrestricted access is not assumed.
Outbound application: proposed	Arrival/window-risk and unloading-duration models, trip agents, provisional reservation service, feasibility checks and planner review. Other departments' models remain unspecified.
Release and execution	Only a planner-approved version enters the TMS and driver workflow. Publishing a tentative option does not change picking, charging, a store promise or actual dispatch.

Note. Current landscape: Company dossier (2026), §A5; Board card C11 (2026), pp. 1–2. Proposed services are design choices. Routiq is the fictional case vendor; PTV is not established as its supplier.

Data ownership and access

Outbound owns route plans, model purpose and operational approvals. Orderpicking owns its load/readiness response. Inventory & SCM agrees the required-versus-accepted quantity contract for OTIF. Storage owns energy/capacity constraints. IT administers technical access, operations and recovery. Incoming transport supplies the company Compliance File; data-use and retention requirements await that review.

Use delivery, route, store, site, vehicle and plan-version keys. Every event needs source time, receipt time, units, correction/version semantics and a validity period. Store data access by purpose and department. Use service permissions that separate read, propose and operational release. Do not expose unnecessary driver identity or customer data.

Capacity and maintenance

Size CPU, memory, storage and resilience against measured model size, training time, concurrent inference, event rate, retention and normal/peak demand. No GPU, server specification or price is justified yet. Prioritise operational inference over scheduled training, isolate workloads and monitor latency/failed jobs. Version models and features; retrain and release after review. Test independent manual TMS operation during platform failure.

4 Coordination and adaptation

A bounded multi-agent planning mechanism — proposal

The design delegates tentative option selection to per-trip agents. A reservation service protects mutually exclusive candidate slots against double claims; it applies a policy fixed by Outbound. It does not centrally optimise every route. Trip agents retain the local choice to request, withdraw and revise their options in response to other agents and new feasibility information.

Table 4
Proposed coordination and adaptation loop

Step	Information, delegated action and feedback
1. Observe	Read current approved TMS version, confirmed constraints, load response and valid predictions. Recompute risk when an input changes.
2. Choose locally	Each trip agent selects a feasible tentative option from its approved option set. Start with conditional 5-minute planning increments for un-departed trips. No real departure moves.
3. Coordinate	Agents exchange candidate requests and window slack. The reservation service accepts a compatible provisional claim under the declared priority/tie rule or returns a conflict.
4. Adapt	A conflicting trip agent releases or revises its request. A changed readiness response invalidates affected claims, which agents re-evaluate. Stop after a bounded negotiation limit and escalate unresolved conflict.
5. Review and release	Show the mutually consistent candidate set, uncertainty and affected stops to the planner. Only an approved plan version can reach the TMS/driver.
6. Evaluate	Log proposals, conflicts, adaptation, human decisions and outcomes. Model or policy changes require a separate human release.

Note. Design inference informed by Gerrits et al. (2024) and Lesson 6 (2026), slides 10–11 and 20–25. Option bounds, priority, tie-breaking, expiry and negotiation limits must be approved and tested before use.

Illustrative test, not an observed FreshWay event

Two trip agents request the same single earlier candidate slot. One receives a provisional reservation under the approved priority rule. The other revises its option instead of retaining a conflicting plan. If the first load becomes unavailable, that reservation expires and both agents adapt again. The planner then decides whether to change either real departure. The test succeeds only if logs show these delegated choices and feedback.

A provisional slot is a planning construct, not an acquired warehouse dock or charging reservation. Orderpicking can initially return feasibility manually; automatic warehouse scheduling needs its own agreement. The shared platform enables this exchange, but does not grant Outbound authority over another department.

Predictions, alerts and shared hosting alone do not satisfy the proposed self-organisation mechanism. Shadow demonstration comes before any operational claim.

5 Decision rights and cross-delivery

Table 5
Proposed decision rights

Decision	Owner and boundary
Allocate / release provisional candidate claims	System, under an approved Outbound coordination policy and a single plan version. No operational release rights.
Actual departure, route or stop-order change	Outbound planner approves every change, including a 5-minute shift. Check all affected stops and record the decision.
Picking priority / charging allocation	Responsible Orderpicking / Storage humans or their separately approved systems. Outbound requests feasibility only.
Store message / driver instruction / subcontract booking	Planner or authorised human, via an agreed channel. No autonomous external communication.
Model / policy / budget / platform rollout	Responsible Outbound/IT owners and Board according to authority. Team direction is not a logged funding decision.

Note. Preserves submitted SA4 (FreshWay Outbound Transport Team, 2026b, pp. 3–7). All new decision delegation is proposed.

Limits, escalation and reclaim

Permit only options that use valid, fresh data and verified store, vehicle, driver, load and energy constraints. Unknown restrictions block the affected option. Missing keys, plan-version conflicts, low-confidence inputs, stale data or unresolved negotiation cause escalation. No change can quietly move another department’s work or a contractual window.

The planner can disable coordination, clear tentative reservations and continue from the last approved TMS plan with a manual review queue. Log who disabled it, invalidated candidates and the reason. Test this recovery before live use. IT can isolate the service without taking over operational decision authority.

Cross-delivery — prepared for Orderpicking

Outbound offers Orderpicking a versioned departure-feasibility request and the proposed system-decision boundaries. The request identifies delivery/trip, load reference, current approved departure, candidate time, reason, source timestamp, plan version and expiry. Orderpicking uses it to return confirmed load availability and whether the candidate is feasible. It retains control over picking priorities.

The platform uses that response to withdraw or revise conflicting provisional options before the planner releases any actual change. Agree keys, response time, correction handling, owner and escalation in the Interface Agreement. This request/response contract is proposed, not an existing live feed.

Receiver confirmation: pending. Orderpicking must write its own line confirming receipt and usability. No Interface Agreement signature, handover date or approval has been invented.

Next commitment: Sander takes this payload and decision policy to the next available Data Council/MT discussion, obtains a dated receiver response and records the agreed follow-up date. If it is unavailable, document the assumption and continue desk/shadow design without treating the interface as accepted.

6 Stakeholder impact map

Technology → behaviour → effect → value. All effects below are hypotheses to test. A shared platform enables the exchanges; it does not by itself establish autonomy, benefits or other departments' requirements.

Table 6
Stakeholder impact map, part 1

Stakeholder	Technology	Behaviour	Effect to test	Value and tradeoff
Outbound planners	Risk/unloading models and trip agents	Review a consistent option set; approve each real change; log rejection reasons	Test earlier useful warnings and fewer conflicting proposals	Service and usable planning time may improve. Outbound carries review/training work; false alerts can increase it.
Own-fleet drivers	Versioned approved TMS instructions	Acknowledge and execute approved plans; report actual events	Test fewer late instruction conflicts and safe feasible shifts	Potentially less time pressure. Driver availability/rest remain binding; no headcount saving assumed.
Subcontractor	Contract and vehicle checks	Planner checks booking cutoffs, access and liability before proposing a trip change	Reject infeasible subcontract options	Avoidable disputes may fall. C03's diesel/access/fine ambiguity remains a human contract issue.
Store receiving teams	Approved ETA range and unload estimate	Plan receiving staff around confirmed feasible arrivals; report local restrictions	Test punctuality, early arrivals, waiting and unloading	Potentially more reliable receipts, with possible earlier staff work. Preserve non-ZE service.
Customers / residents	Window, access and noise constraints	Store receives within permissions; planner rejects nuisance-producing options	Monitor complaints and possible shelf-availability effect	Customers may gain availability; residents avoid extra nuisance. Punctuality alone proves neither.
Orderpicking	Agreed feasibility request/response	Confirm actual load readiness and candidate feasibility; retain priority authority	Test complete readiness data and feasible handovers	Potentially fewer missed handovers. Data/response effort and disrupted warehouse work must be counted.

Note. Authors' proposed impact map. Sources: Company dossier (2026), §§A2/A4/A5; department files; Board cards C03, C05, C10, C11 and C12 (2026); Study Guide v2 interface/constraint rules; submitted SA4. External concept: Gerrits et al. (2024).

The fixed course counterparties are Orderpicking, Inventory & SCM and Delivery to consumers. Storage and Incoming transport add company-wide constraint dependencies. Other departments must state their own actual requirements; access to a shared server does not imply access to all data or authority over shared resources.

6 Stakeholder impact map — continued

Technology → behaviour → effect → value. All effects below are hypotheses to test. A shared platform enables the exchanges; it does not by itself establish autonomy, benefits or other departments' requirements.

Table 7
Stakeholder impact map, part 2

Stakeholder	Technology	Behaviour	Effect to test	Value and tradeoff
Inventory & SCM	Delivery-keyed re quired/accepted quantities	Reconcile quantities and timing using agreed keys/corrections	Measure join coverage and eventual true OTIF	Better service/stock evidence; SCM bears reconciliation work. Missing POD cannot be replaced by GPS.
Delivery to consumers	Shared-constrain t consultation	State relevant city-access, energy or resource conflicts through an agreed exchange	Identify spillovers before an Outbound change	Protect consumer-delivery execution. No shared fleet, van-feed or model requirements assumed.
Storage / facilities	Energy & Capacity constraint input	Confirm availability limits; retain charging control	Reject plans relying on unavailable energy/vehicles	Feasible departures without displacing cold-chain work; Storage bears checks. Smart charging is separate scope.
Incoming / compliance	Compliance File and data-use requirements	Clarify acceptable data use and transport-information controls	Track unresolved deployment dependencies	Protect company and data subjects. Outbound cannot claim a compliance approval that was not supplied.
Central IT / data team	Shared hosting, permissions and monitoring	Isolate workloads; schedule training; monitor latency and recovery	Test reuse, queue delays, availability and support effort	Possible shared support value. IT bears operations/integration costs and concentrated outage risk.
Board / Finance / environment	Benefit/cost register and measured km/energy	Stage investment; allocate costs; compare service, cost and environmental outcomes	Test total cost and km/energy per delivered cage	Value may fall in stores/Outbound while IT pays. Environmental benefit is conditional; detours/idling can worsen it.

Note. Authors' proposed impact map. Sources: Company dossier (2026), §§A2/A4/A5; department files; Board cards C03, C05, C10, C11 and C12 (2026); Study Guide v2 interface/constraint rules; submitted SA4. External concept: Gerrits et al. (2024).

The fixed course counterparties are Orderpicking, Inventory & SCM and Delivery to consumers. Storage and Incoming transport add company-wide constraint dependencies. Other departments must state their own actual requirements; access to a shared server does not imply access to all data or authority over shared resources.

7 KPIs and evaluation

Table 8
KPI references and evaluation

Measure	Reference and evaluation
Arrival service	Retain supplied ZE 87.47%/other 98.38% as historical flags. Confirm the boundary rule and measure inside-window, early and late outcomes consistently for intervention and comparison groups.
Forecast quality	No accuracy result exists. Compare ETA and unloading MAE separately against simple held-out medians/current estimates. Evaluate risk calibration and useful alerts at an agreed threshold.
OTIF	Case 97.2%, company target 98.5%. Calculate only after required-versus-accepted quantities/POD and timing semantics are available at delivery grain.
Cost per delivered cage	Case ~EUR 7.73, company target EUR 7.00. Sample allocated ratio EUR 7.1237 = EUR 3,701,054.56 / 519,540 cages. Reconcile scope before comparing or monetising gains.
Utilisation	Case 45.4 cages/route, target 52. Sample 45.13; route-specific weighted capacity utilisation 73.19%. Use actual 66/40/30-cage capacities, not 66 for every vehicle.
Human and platform effects	Measure planner minutes, false alerts, overrides, negotiation/conflicts, readiness-response effort, latency, uptime, incident/recovery time and other-store spillovers. No measured workload reduction exists.

Note. Case references: Department file: Outbound transport (2026), pp. 2–3; sample calculations: store_replenishment.csv and vehicles.csv. Code and calculation evidence accompany the project pack.

Comparison and attribution

Compare current approved planning, simple statistical estimates with identical approval rules, and AI predictions with the proposed bounded coordination. Keep prediction horizons, constraints and data availability comparable. This separates the value of earlier scheduling, integration, predictions and coordination.

Split training and evaluation chronologically, keeping whole route-days together. Fit medians/preprocessing on training data only. Evaluate ZE/non-ZE, site, vehicle type and peak periods separately. Do not use future actual arrivals, the same stop’s actual unloading, final driving duration, costs or energy as pre-dispatch features.

For live evaluation, use matched or randomised route-days where feasible. Report counts, percentage-point change and uncertainty that respects route clustering. Agree a useful service improvement and acceptable cost, early-arrival, safety and other-store limits before results. A proposed four-week pilot may need extension for enough evidence. No binding 95% ZE norm, MAE 5-minute requirement or 2% early-arrival limit is established by the case.

8 Implementation, costs and risk

Table 9
Implementation gates

Gate	Required result before progressing
A. Requirements and data	Confirm appointment/flag rules, offset rationale, unloading label and inconsistent stop sequences. Agree required data keys and prospective logging. Collect each department's requirements without inventing applications.
B. Shadow coordination	Benchmark simple/AI forecasts. Demonstrate competing trip-agent requests, adaptation and deterministic conflict handling without dispatch changes. Test permissions, expiry, contention and manual recovery.
C. Limited live pilot	Only after Board/operational approval, priced interfaces, usable recipient agreements and store/vehicle/load/driver/energy checks. Planner approves each actual revision.
D. Scale or stop	Compare service, costs, workload and network guardrails. Expand only on adequate evidence. Record inconclusive or negative findings and retain a viable manual process.

New board-card implications

C10 identifies 20 of 22 ZE stores without trailer-compatible docks and 37 stores with pre-07:00 noise conditions. The affected permit IDs are not all identified; store-level checks remain necessary. C03 gives diesel-only subcontract store capacity, booking deadlines and unresolved ZE fine liability. Neither card explains the constant 15-minute offset, and an AI model cannot resolve vehicle-access or contract gaps.

C11 places the TMS in the vendor cloud with paid position export and a 31 March 2027 contract end. WMS notice is due 31 December 2026; the BI cloud roadmap also matters. The platform must fit these decisions rather than assume a new departmental TMS. C09 requires peak-volume testing, interface pricing before purchase, Storage checks and operating continuity. Do not infer a safe MW charging budget by subtracting MW loads from an MVA contract without the appropriate electrical basis.

Costs, benefits and safeguards

Obtain incremental quotations for configuration, API access, integration, hosting, security, maintenance, training, monitoring and resilience. Do not count the existing EUR 0.45 million annual TMS contract as a new AI investment. Size hardware only after ordinary/peak benchmarks. Outbound owns its model and pilot work; IT operates shared infrastructure. Finance must agree a cost allocation method and avoid double-counting savings.

Potential value includes shared support and earlier useful planning decisions. Risks include common outages, resource contention, inaccurate forecasts, sensitive location data and additional review/warehouse/store work. Limit inference permissions, show data age and uncertainty, retain approved versions and test manual fallback. No equipment purchase, fleet sharing, charging automation, job reduction or net saving is assumed.

Sources: Board cards C03, C05, C09, C10, C11 and C12 (2026); Company dossier (2026), §A4. All approvals and procurement remain separate from this design recommendation.

9 Evidence appendix and submission checks

Table 10
Rubric and evidence map

Requirement / criterion	Evidence in this SA5 package
Recent academic article and concept	Report §2; Gerrits et al. (2024), full-text review and verified DOI. Concept/deployment distinction explicit.
Data and traceable analysis	Report §§1,7; raw-source selection, field definitions and reproducible analysis script/metrics. Supports rubric criteria 2 and 4.
Technology and integrated solution	Report §§3–5,8; current/proposed architecture, bounded coordination, requirements, costs/risks. Supports criteria 1 and 3.
Collaboration: criterion 5	Report §5 cross-delivery and §6 map. Receiving department's usable-receipt response and signed agreements remain pending. Agent reviews do not substitute for interdepartment collaboration.
Presentation: criterion 6	Three-slide deck with architecture, impact map and decision/evaluation. Detailed notes contain sources.
Reflection: criterion 7	The human team's personal/group reflection belongs to the final portfolio. No personal learning or attendance is invented here.

Note. Rubric map for this sub-assignment, not a claim that all final-portfolio criteria have been completed. Source: Study Guide v2; Rubric Smart Logistics (2026); Before you hand in (2026).

Contribution provenance and corrections

Quinten's named research/concept document and Francesco's attributed Section3 report are present. Two SA5 decks provide technical/impact material, but their filenames alone do not establish Made's or Scholte's authorship or completion. Sander's integration has now been prepared. The team must confirm contribution attribution; no attendance or effort judgement follows from missing labels.

Assembly corrects the tier definitions, unverified deployment/approval language, autonomous charging scope, unsupported workload 45 to 10 minutes and guaranteed savings/trip-reduction claims. It preserves the supplied research direction, impact material and evaluation ideas. An unverified Schulte/Voss citation is excluded; the checked Gerrits review carries the academic requirement.

Remaining human completion actions

Check the actual Brightspace deadline/status and submit the report plus three-slide presentation. Confirm Made's preferred spelling against the roster. Obtain Orderpicking's own receipt/usability line, record the next commitment in MT minutes and keep the Board decision reference when one exists. Complete the required personal AI declaration only after each author has actually reviewed and revised the work.

These are evidence/action gaps. No sender message, receiver signature, Board approval or Brightspace upload has been performed by preparing these files.

10 AI declaration and responsibility

Actual use recorded for this preparation

During preparation, ChatGPT Work was used to inventory and extract the supplied project files, compare course and previous-assignment requirements, check sources, recalculate descriptive data, identify unsupported claims, design the agent roles and bounded coordination proposal, and assemble the report, Board Memo and presentation. Five specialist AI workers reviewed evidence, research, data, architecture and stakeholder impacts. Assistance was substantial, including new draft text and visual/document production.

The supplied contributions also disclose earlier Claude and Gemini Notebook assistance. Tool versions for that earlier work are not verified. No model training, live logistics pilot, deployed integration, recipient confirmation or completed personal review is claimed by this declaration.

Human responsibility

The team's vision and submitted work provide the project direction and source material. The five named students remain responsible for verifying the resulting claims and sources, revising the text into work they understand and can defend, and checking the final submission. Preparing this report does not establish that each student has already completed those actions.

The study guide permits AI assistance but requires human input, editing and responsibility. The standard declaration below is supplied for human completion; its review attestation must not be signed until it is true (Study Guide Smart logistics, 2026, "Use of generative AI").

Windesheim declaration — for authors' completion after review

"During the preparation of this work, I used ..... [name of the tool + version number, if applicable] to ..... [rephrase my own sentences/sections of text, summarize my own sections of text, structure my sections of text, start writing, generate ideas, ..]"

"After using this tool, I thoroughly modified the outcomes to ensure that my work reflects my own competencies and learning outcomes. I take full responsibility for the content of this work. I am aware that any actions or omissions intended to impede or result in an incorrect judgment of my knowledge, understanding, and skills may be considered fraud. I am also aware that the literal copying of text/code, etc., from AI obstructs a correct assessment of my knowledge and skills and will be considered fraud."

The first paragraph above records actual assistance. The quoted institutional template is not a completed personal attestation. Each author must supply the accurate tool/use details and complete the review before submitting it as their declaration.

AI reference: OpenAI. (2026). ChatGPT [Large language model]. https://chatgpt.com/ . The service name is verified; an exact model/version for all assistant work is not recorded. No agency label is used to imply a level of human review that has not happened.

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Internal documents without a credited author use the title in the author position. Board-card dates use supplied release footers; no dataset publication date is invented. Course artefacts are accessible to the assessors through the supplied project environment. The peer-reviewed article's first online publication was December 2023; this reference uses its 2024 volume/issue.

Source checks and calculations prepared 4 November 2026. The repository's partial SA4 contributions were compared with the actual submitted SA4 report/presentation before integration. Unverified claims in individual contributions were not carried forward as facts.